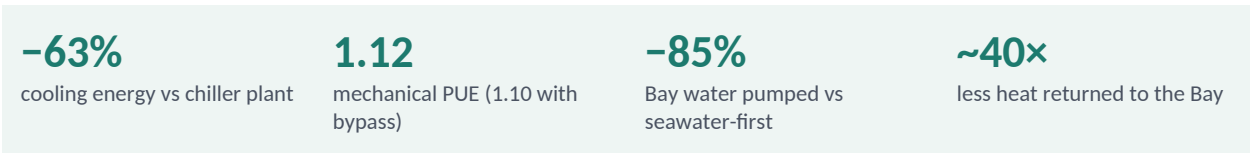


Does the Breeze Save Energy? Fan, Pump and Chiller Power for a Hybrid-Cooled Pier Data Center

Vishnu Sai Sharan Ankathi • Independent research • September 2026 • Python energy model, 3 years of hourly data



1. Summary

The heat exchanger analysis showed the breeze can hold 21 °C supply air in 84.8% of hours. This report asks the next question: once every fan and pump is counted, does the hybrid design actually use less electricity, and compared with what?

Result: the hybrid uses about 275 MWh of cooling electricity per year for a 270 kW room (mechanical PUE 1.12), about two-thirds less than an air-cooled chiller plant. But a design that pumps Bay water every hour uses less still (157 MWh, PUE 1.07), because fans cost more than pumps. The hybrid's real advantage is the Bay: it pumps about 85% less seawater and returns roughly 40 times less heat. The design is an energy-environment trade-off, not an energy win over seawater.

2. What Uses Power

With the chiller almost never running, cooling electricity comes from three places:

Consumer	What it does	When it runs	Design power
Room-air fans	Move 10.06 m³/s per path through ducts, both exchangers and the room	24/7	11.4 kW per path
Outdoor-air fans	Pull marine air through filters and the Stage 1 exchanger	Variable speed, only as fast as needed	9.2 kW per path at full speed
Seawater pump	Push Bay water through the Stage 2 coil	Only when the breeze falls short	0.85 kW per path
Chiller	Trims whatever is left	0.14% of hours	COP 5

Fan and pump power follow $P = \text{airflow} \times \text{pressure drop} \div \text{efficiency}$. Outdoor-air fans slow down in cold weather, and because fan power scales with speed cubed, halving the speed cuts power by about 88%.

3. Method

3.1 Assumptions

Item	Value	Basis
Room-air loop pressure drop	680 Pa	Ducts 250, Stage 1 200, seawater coil 120, containment/filter 110
Outdoor-air loop pressure drop	550 Pa at full flow	MERV 13 filters 150, intake 100, Stage 1 200, exhaust 100
Fan + motor + drive efficiency	0.60	Typical packaged air handler
Outdoor fan minimum speed	20%	Typical VFD lower limit
Seawater pump	160 kPa, 60% efficient	~5 m lift to the deck, coil, strainer, piping
Electricity price	\$0.125/kWh	US industrial average

These are typical design values, not vendor data. They are the main uncertainty and are varied in Section 5.

3.2 Options compared

- **Chiller only:** an air-cooled chiller (COP 4) and chilled-water coil every hour. The conventional baseline.
- **Seawater first:** Bay water through the coil every hour, chiller trim. The conventional way to use the Bay, as in existing water-cooled designs.
- **Hybrid (breeze first):** this project's design.
- **Hybrid + coil bypass:** room air skips the dry seawater coil whenever the pump is off.

Mechanical PUE = (IT load + cooling electricity) ÷ IT load. UPS and lighting losses are left out because they are the same for every option.

4. Results

4.1 Annual cooling electricity

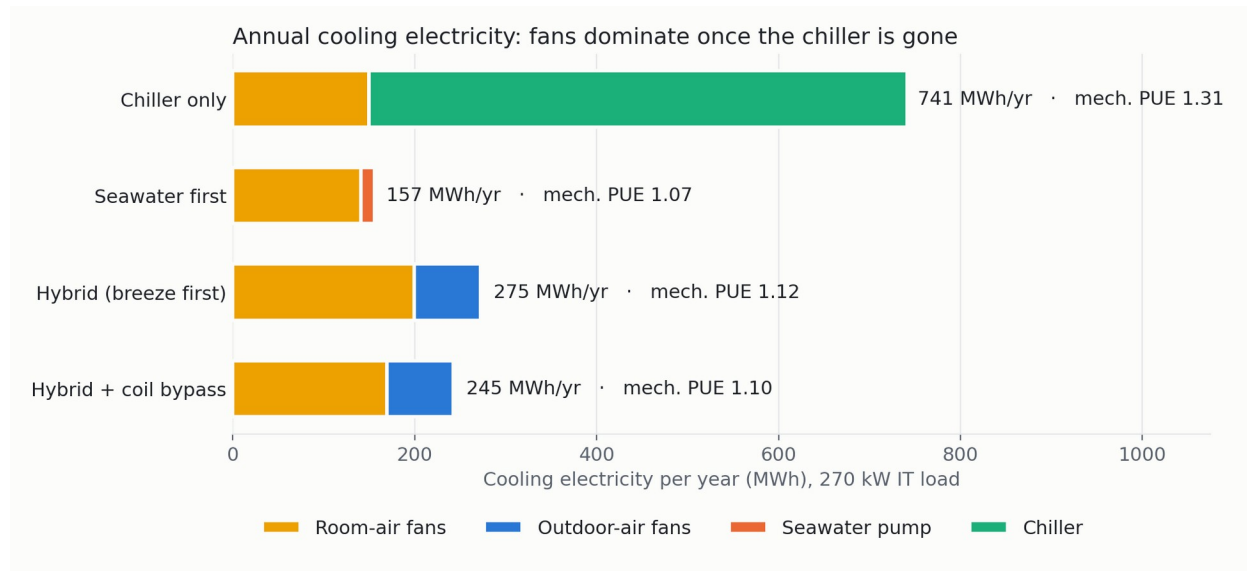


Figure 1. Annual cooling electricity by component for a 270 kW IT load at 21 °C supply.

Option	Cooling electricity	Mech. PUE	Electricity cost	vs chiller only
Chiller only	741 MWh/yr	1.31	\$92,600/yr	—
Seawater first	157 MWh/yr	1.07	\$19,600/yr	–79%
Hybrid (breeze first)	275 MWh/yr	1.12	\$34,300/yr	–63%
Hybrid + coil bypass	245 MWh/yr	1.10	\$30,600/yr	–67%

Against a more efficient chiller plant (COP 6, 544 MWh/yr), the hybrid still saves about 50–55%. Once the chiller is gone, the room-air fans become the largest single consumer, about 70% of the hybrid's cooling energy, because they run all year against every exchanger in the loop.

4.2 Why seawater first uses less energy

Water carries about 3,500 times more heat per unit volume than air, so moving heat with a pump takes far less power than moving it with a fan. The seawater pump draws 0.85 kW per path; pulling outdoor air through filters and an exchanger takes about 9 kW per path at full speed, and the air-to-air exchanger adds another 200 Pa to the room-air loop. On energy alone, the breeze does not beat the Bay.

4.3 Bay water and heat discharge

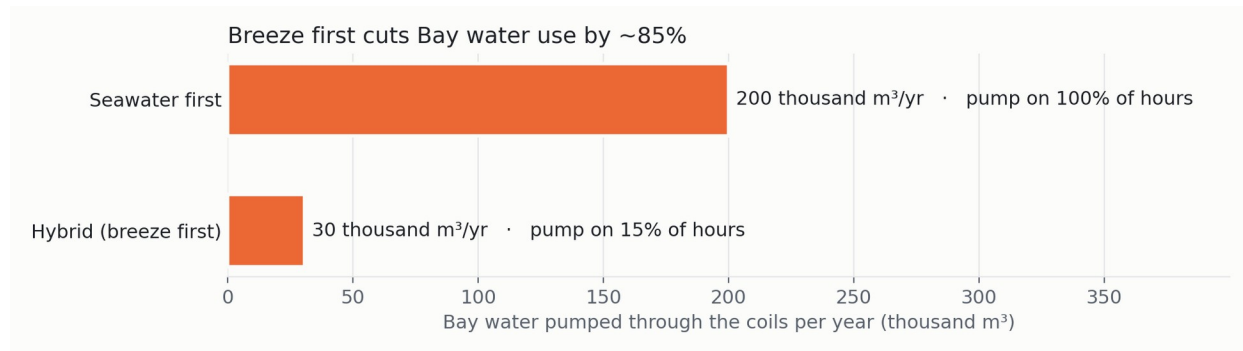


Figure 2. Bay water pumped through the coils per year.

Bay impact (per year)	Seawater first	Hybrid	Change
Seawater pumped	200,000 m³	30,500 m³	~85%
Pump running	100% of hours	15.2% of hours	~85%
Heat returned to the Bay	~2.4 GWh thermal	~0.06 GWh thermal	~40× less

The hybrid's extra 118 MWh over seawater first costs about \$14,700 a year, roughly \$0.09 for every cubic metre of Bay water not pumped. Continuous intake and thermal discharge are exactly what led to the 2019 rejection of a pier-mounted, water-cooled data center at Alameda Point.

5. Sensitivity

Case	Cooling electricity	Mech. PUE	Pump runtime
Baseline (ε 0.75, airflow ratio 1.0, 21 °C)	275 MWh/yr	1.12	15.2%
Supply 24 °C	243 MWh/yr	1.10	4.4%
Outdoor-air ratio 1.5	249 MWh/yr	1.11	6.7%
Stage 1 ε 0.65	309 MWh/yr	1.13	36.9%
Stage 1 ε 0.85	257 MWh/yr	1.11	7.0%
All pressure drops +30%	356 MWh/yr	1.15	15.2%
All pressure drops -30%	193 MWh/yr	1.08	15.2%
Fan efficiency 0.50 / 0.70	329 / 236 MWh/yr	1.14 / 1.10	15.2%

A 24 °C supply is the strongest lever: it lowers energy and cuts seawater use by about 70% at the same time, because the outdoor-air fans can run slower. This assumes the rack temperature rise stays at 12.25 K (return air 36.25 °C), which CFD at 24 °C still has to confirm. Pressure drop is the biggest uncertainty; even at +30% the hybrid stays about half the chiller plant's energy.

6. Findings

- **The hybrid cuts cooling energy by about two-thirds** compared with a chiller plant (mechanical PUE 1.10–1.12).

- **Seawater first would use less energy.** Pumps beat fans. The breeze-first design should not be sold as an energy win over seawater.
- **The hybrid's advantage is Bay impact:** about 85% less seawater pumped and about 40 times less heat returned, for about \$0.09 of electricity per m³ avoided.
- **Pressure drop is the design driver.** Low-resistance exchangers, short ducts and a seawater-coil bypass matter more than any control tweak.
- **Use a 24 °C supply if the servers allow it:** lower energy and pump runtime about 4%, once CFD confirms rack inlet temperatures.

7. Limitations

- Pressure drops and efficiencies are typical values, not vendor data.
- The heat balance uses the CFD return air temperature (33.25 °C), which implies about 292 kW of heat removal against 270 kW of IT load, so results are slightly conservative. Fan heat added to the room air is not yet included.
- One weather point and one tide station; water data after February 2024 filled with averages.
- Environmental benefits (entrainment, thermal plume) are estimated from volumes and heat, not modeled.

8. Next Steps

- Size the Stage 1 air-to-air exchanger (plate area, volume, pressure drop) and check it fits on the pier.
- Replace assumed pressure drops with vendor data; add fan heat and coil condensation.
- CFD at 24 °C supply to confirm rack inlet temperatures and the return-air temperature used in the 24 °C case.

Code and data: github.com/sharanankathi/bay-breeze-cooling (energy_model.py). Data: Open-Meteo (CC BY 4.0) and NOAA CO-OPS (public domain).

Research idea, design decisions and modeling direction by the author. AI tools were used to help write code, run calculations and draft this report.